

A geometry-constrained rock–TBM spatial interaction graph for component-indexed anomalous interaction fields in TBM excavation

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Abstract

TBM excavation responses are shaped by changing spatial relations between the advancing machine and surrounding rock. Monitoring curves and geological voxel fields, however, do not specify which geological units are plausible interaction partners for individual TBM components. This study proposes a geometry-constrained rock–TBM interaction graph computation method for representing component-indexed anomalous interaction fields. At each excavation step, TSP-derived geological voxels and component-labelled TBM surface nodes are organised in a common excavation coordinate, and candidate rock–machine edges are screened by active-zone, distance, normal-compatibility, and excavation-state constraints. The resulting sparse candidate-edge set is aggregated by TBM component to compute a component $\times$ excavation-step field, including interaction intensity $I_c(t)$ and step-level change $\Delta I_c(t)$. The field is aligned with monitoring responses and queried back to contributing rock–machine edges, providing a computable spatial basis for excavation anomaly readout. Two tunnel cases show that the graph computation generates component-level anomaly fields that can be compared with monitoring responses; the clearer SJLS interval exhibits stronger residual-consistent co-variation than the BSLL intervals. The proposed computation is therefore treated as a queryable summary of screened candidate relations, not as measured contact force, calibrated jamming probability, or a universal operational risk rule.

Keywords: spatial representation; rock–machine interaction graph; component $\times$ excavation-step anomalous interaction field; geometry-constrained candidate relation; persistence residual; excavation anomaly discovery; jamming-risk-oriented interpretation; TBM excavation

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1 Introduction

TBM excavation is a moving spatial interaction process, not a fixed-site monitoring problem. As the cutterhead and shield advance, geological units enter, approach, and leave the machine’s potential interaction zone. Changes in thrust, torque, penetration, advance rate, and shield pressure therefore reflect both operational control and the evolving spatial configuration between geological anomalies and machine components.

Monitoring-based TBM response models are useful because short-horizon excavation responses often have strong temporal persistence. Tabular and sequence-based models learn this persistence from recent operational and response histories [2, 12, 13, 35]. However, a monitoring curve does not state which geological unit is adjacent to the cutterhead or shield, whether the

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adjacency is geometrically plausible, or how the candidate rock-machine relation set changes as excavation proceeds. This limits the spatial interpretability of monitoring-only analyses.

Advance geological prediction and TSP surveys provide the complementary view: they describe spatial variation in the ground ahead of or around excavation [18]. These data can be converted into chainage-referenced geological voxels or attribute fields, preserving where low-velocity or other geological anomalies occur. Yet a geological voxel field by itself answers where an anomaly is located, not which TBM component is a plausible interaction partner at a particular excavation step. A component-indexed field representation must therefore link geological spatial support to machine surface geometry and excavation state.

To address this gap, the paper treats rock-TBM interpretation as a spatial entity-relation modelling problem. The input consists of a TSP-derived rock voxel field, an advancing parameterized TBM surface, and excavation-step-aligned monitoring records. Two graph entity types are defined: rock voxels and TBM surface nodes. Monitoring responses are aligned to excavation steps but are not graph entities. The method then constructs a geometry-constrained sparse rock-TBM interaction graph at each excavation step using active-zone, distance, normal-compatibility, and excavation-state constraints. Weighted rock-machine edges are aggregated by TBM component to compute a component $\times$ excavation-step anomalous interaction field. The field contains $I_c(t)$ and $\Delta I_c(t)$, can be aligned with monitoring responses, and remains traceable back to individual rock-machine edges. The graph is introduced first as a spatial data model and graph computation method, not as a predictive graph-learning architecture.

The contribution lies in integrating these elements into one excavation-step-indexed spatial graph computation, rather than in claiming a new high-accuracy predictor. The paper contributes: (i) a rock-TBM spatial entity model that places TSP-derived rock voxels and TBM surface components in a common excavation coordinate while keeping monitoring responses as aligned records; (ii) a per-step geometry-constrained sparse rock-machine interaction graph built from active-zone membership, distance decay, surface-normal compatibility, and excavation-state constraints; and (iii) a component $\times$ excavation-step anomalous interaction field that aggregates fixed-reference geological anomaly scores over screened candidate relations per TBM component and tracks $I_c(t)$ and $\Delta I_c(t)$ over excavation steps. The empirical evidence is used to support representation, traceability, and selected-interval residual consistency, not operational fault attribution or forecasting superiority.

2 Related work

2.1 TBM response modelling and spatial gaps

Traditional TBM performance studies estimate penetration, advance rate, thrust, torque, or related operational responses from rock mass properties and machine parameters. Empirical, statistical, and machine-learning formulations are used for planning and performance assessment in hard-rock tunnelling [2, 12, 35]. With the accumulation of monitoring data, sequence models such as LSTM and GRU also learn response inertia and lagged dependence from recent operational histories [5, 13]. These studies provide important response-prediction baselines, but their spatial unit is usually a tunnel section, a sample, or a feature vector rather than an explicit relation between a geological object and a TBM component.

This limitation is central to the present paper. A monitoring curve can describe how torque, thrust, penetration, advance rate, or shield pressure changes with time or chainage, but it does not state which geological voxels are plausible interaction partners for the cutterhead, front shield, middle shield, or tail shield. The present study therefore does not try to replace monitoring-based predictors. It asks whether a spatial relation representation provides component-indexed field information about the rock-machine configuration that monitoring curves alone do not encode.

2.2 Geological prediction as spatial support

Advance geological prediction methods, including TSP and related geophysical or drilling-based approaches, are used to identify faults, fractured zones, water-rich zones, weak surrounding rock, and other geological anomalies before or during tunnel excavation [18]. For TBM analysis, such data are valuable because they describe spatial heterogeneity that is not contained in the monitoring record itself. TSP results can be represented as velocity profiles, sections, or voxelised attribute fields along the tunnel direction; voxelisation is useful here because it preserves spatial support, local attribute values, and chainage-referenced geological context.

Geological spatial representation alone is nevertheless incomplete for the diagnostic question addressed in this paper. A voxel field can show where anomalous ground occurs, but it does not specify whether that anomaly is adjacent to the cutterhead or shield, whether the adjacency is normal-compatible, or whether the voxel remains active at the current excavation step. This is a spatial-support problem as much as an engineering problem: TSP voxels, TBM surface samples, and monitoring records have different spatial or observational supports and cannot be interpreted as a single tabular unit without losing relation information [11, 21, 22].

2.3 Entity–relation spatial data models

GIScience is concerned not only with mapping locations but also with representing geographical objects, spatial relations, temporal change, and processes [9, 11, 30]. Spatial relation theory provides concepts for neighbourhood, adjacency, containment, intersection, and other topological or metric relations [7, 31]. TBM excavation fits this perspective because rock units, machine surfaces, and monitoring responses are observed in a moving excavation coordinate, and the relevant relations change as the machine advances.

Recent geospatial knowledge-graph and graph data-model studies show how heterogeneous spatial entities are integrated for retrieval, reasoning, and query [17, 25, 34, 40]. Scenario-oriented spatiotemporal knowledge-graph models likewise emphasise linking heterogeneous data through meaningful objects and relations rather than storing them only as parallel indicator tables [28]. The present paper shares this entity–relation modelling perspective, but it operates at a finer voxel and surface-node support and targets a moving underground engineering process rather than an indoor social network, an urban knowledge service, or a regional knowledge base.

This literature provides the conceptual foundation for the proposed graph: rock voxels are geological objects, TBM surface samples are machine-side spatial objects, and rock–machine edges encode screened candidate interaction relations. Monitoring responses are kept as aligned records rather than graph nodes. The representational gap is therefore not the absence of a predictor, but the lack of a spatial data model that connects geological voxels, machine components, candidate interaction relations, and response residuals in one component-indexed field representation.

2.4 Spatial interaction graphs

Graphs are a natural formalism for representing irregular spatial objects and their relations. Recent geospatial studies have used graph-based representations for spatial classification, spatial multivariate distributions, urban network morphology, and spatially constrained community detection [4, 6, 20, 39]. Spatial interaction models further emphasise that relations are shaped by distance decay, compatibility, direction, accessibility, and spatial support [8, 29]. For the rock–TBM problem, this means that candidate relations should not be fully connected or purely learned without physical constraints. They should depend on active-zone membership, distance, surface-normal compatibility, and excavation state.

Graph neural networks and heterogeneous graph models remain relevant as future extensions for learning on structured data [27, 32, 36, 38]. Spatiotemporal graph models have also been

successful in traffic, flow prediction, and physical simulation [10, 19, 24, 26, 33, 37]. However, these studies mainly motivate the representational capacity of graph structures and the need for spatially constrained relations. They do not imply that the present tunnel cases should be framed primarily as a black-box graph-learning benchmark.

In the tunnelling domain, digital twin frameworks for TBM performance prediction, visualisation, monitoring, and deformation prediction have been proposed [1, 14, 16]. The present graph-based representation is complementary: rather than replacing prediction-oriented digital twin modules, it specifies an underground spatial relation layer between geological voxels and TBM surface components. This layer defines active candidate relations, component-indexed aggregation, and edge-level traceability, which are typically implicit in monitoring-centred digital twin workflows. The evaluation is therefore kept descriptive: Spearman association, detrending, null comparisons, sensitivity analysis, and edge-level traceability are used to support a spatial field-representation claim, not a universal prediction or risk-calibration claim.

3 Methodology

This section formalises rock–TBM interaction as an excavation-step-indexed spatial entity-relation graph. The method starts from a TSP-derived rock voxel field, an advancing parameterized TBM surface, and aligned monitoring records under a common excavation-step index. It then constructs a sparse rock–TBM interaction graph at each excavation step using active-zone, distance, normal-compatibility, and excavation-state constraints. Finally, weighted rock–machine edges are aggregated by TBM component to compute a component $\times$ excavation-step anomalous interaction field. This field supports excavation anomaly indication, component-indexed readout, and jamming-risk-oriented interpretation while retaining traceable edge contributions. The graph is sparse because rock–machine edges are not fully connected; they are retained only when the four spatial and excavation-state constraints are jointly satisfied. Figure 1 summarises the method structure.

The three subsections correspond to the input representation and two core algorithmic operations. Section 3.1 defines the two graph entity types and the aligned monitoring records. Section 3.2 defines geometry-constrained sparse graph construction, and Section 3.3 defines component-level anomalous interaction field computation from screened rock–machine edges.

The formulation estimates a queryable field rather than a direct mechanical response. Given geological voxels D_{geo} , a parameterized TBM surface M_{TBM} , and monitoring records aligned by excavation step, the graph-construction operator

$$\mathcal{G} : (D_{\text{geo}}, M_{\text{TBM}}, t) \mapsto G_t$$

produces sparse rock–machine candidate relations, and the field-computation operator

$$\mathcal{F} : (G_t, \phi(g_i), \rho_j) \mapsto \{I_c(t), \Delta I_c(t)\}_{c \in C, t=1, \dots, T}$$

aggregates those relations into a component-indexed anomalous interaction field. The aligned monitoring records are used to read this field against excavation responses; they are not inputs to $\mathcal{G}$ and do not define graph edges.

3.1 Rock, machine, and monitoring data representation

The input data are organised into two graph entity types and aligned monitoring records: rock voxels, TBM surface nodes, and excavation monitoring responses aligned to excavation steps. Monitoring records are not graph nodes; they are used after graph computation to evaluate and interpret the component-level anomalous interaction field.

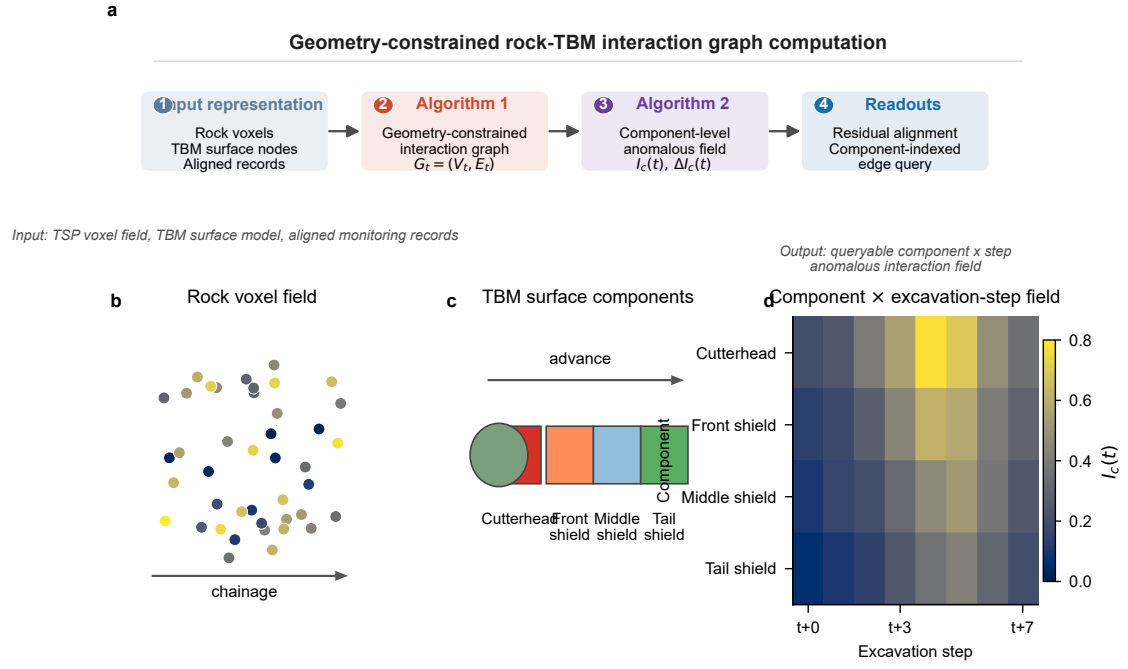


Figure 1: Method overview of the geometry-constrained rock–TBM spatial interaction graph computation. (a) Sparse graph construction from active rock voxels and component-labelled TBM surface nodes using active-zone, distance, normal-compatibility, and excavation-state constraints. (b) TSP-derived rock voxel field with geological attribute support. (c) Parameterized TBM surface components advanced along the excavation direction. (d) Component × excavation-step anomalous interaction field computed by aggregating weighted rock–machine edges by TBM component; the field can subsequently be aligned with monitoring responses and queried back to contributing edges.

Rock voxel. The geological input is a TSP-derived voxel field $D_{\text{geo}} = \{(c_i, g_i)\}$, where $c_i = (x_i, y_i, z_i)$ is the coordinate of rock voxel i and g_i is its geological attribute vector (e.g. P-wave velocity V_p , S-wave velocity V_s , dynamic Young’s modulus, velocity ratio, Poisson’s ratio, density). Rock voxels are the geological spatial support of the graph.

TBM surface node. The machine input is a parameterized TBM surface. Each surface node is represented by $(p_j(0), n_j, \rho_j)$, where $p_j(0)$ is the initial surface-node coordinate, n_j is the local outward surface normal, and ρ_j is the machine component label. The current implementation partitions the machine surface into cutterhead, front shield, middle shield, and tail shield components; ρ_j therefore indexes one of these four labels. TBM surface nodes are the machine-side spatial support of the graph.

Aligned monitoring records. At each excavation step t the monitoring vector is

$$u_t = [\text{AdvanceRate}_t, \text{RPM}_t, \text{Torque}_t, \text{Thrust}_t, \text{Penetration}_t, \text{ShieldPressure}_t],$$

and the response variables used downstream are denoted $r_t^{(k)}$, where k indexes a response channel such as torque, thrust, penetration, advance rate, or shield pressure. Monitoring responses are aligned to the graph output $I_c(t)$ and $\Delta I_c(t)$; they do not participate in graph construction.

Common chainage coordinate. Because TSP coordinates and TBM monitoring mileage are commonly recorded in different local frames, the two graph entity types and the monitoring records are mapped to a common local chainage coordinate. With $\text{ShieldMileage}(t)$ the shield mileage stamp at step t , a simple alignment is

$$x_{\text{local}}(t) = \text{ShieldMileage}(t) - \text{ShieldMileage}_{\text{start}},$$

after which TBM surface nodes are advanced along the excavation direction d :

$$p_j(t+1) = p_j(t) + \Delta x d.$$

Rock voxels, TBM surface nodes, and monitoring records are therefore aligned to the same excavation-step index t . This subsection only establishes that common index; it does not treat monitoring records as graph entities.

Table 1 summarises the resulting entity–relation data model. The table makes explicit that the graph is a queryable entity–relation representation rather than only a plotting device or a neural-network input tensor. Figure 1(b–d) illustrates how the rock voxel field, TBM surface components, and component-indexed field readouts are formalised under the common excavation-step reference.

Table 1: Entities, relations, and readouts in the geometry-constrained rock–TBM interaction graph computation.

Entity or quantity	Definition	Role
V_t^r	Active TSP-derived rock voxels	Geological spatial support
V_t^m	Parameterized TBM surface samples	TBM component support
E_t^{rr}	Rock–rock voxel neighbourhoods	Geological contiguity context
E_t^{mm}	TBM surface neighbourhoods	Machine-surface topology context
E_t^{rm}	Screened rock–machine candidate relations	Candidate interaction support
$A_c(t)$	Sum of geometry weights for component c	Auxiliary exposure readout
$I_c(t)$	Geometry-weighted anomaly intensity for component c	Main field readout
$e_{t+h}^{(k)}$	Persistence residual of response k	Aligned monitoring readout

3.2 Geometry-constrained sparse graph construction

Algorithm 1 summarises the graph construction operation. Its purpose is to produce a sparse and queryable relation support E_t^{rm} for each excavation step, rather than to infer measured contact.

Algorithm 1 Geometry-constrained sparse graph construction

Require: Rock voxels $\{(c_i, g_i)\}$, TBM surface nodes $\{p_j(0), n_j, \rho_j\}$, excavation steps $t = 1, \dots, T$

Ensure: Graph snapshots $G_t = (V_t^r \cup V_t^m, E_t^{rr} \cup E_t^{mm} \cup E_t^{rm})$

```

1: for each excavation step  $t$  do
2:   Advance the TBM surface to  $p_j(t)$  using the common excavation-step coordinate.
3:   Select active rock voxels  $V_t^r$  using the face and shield active zones.
4:   Construct  $V_t^m$ ,  $E_t^{rr}$ , and  $E_t^{mm}$  from voxel and TBM-surface neighbourhoods.
5:   for each active rock voxel  $i$  and TBM surface node  $j$  do
6:     if  $d_{ij}(t) \leq \tau_{\text{edge}}$ ,  $\kappa_{ij}(t) \geq \eta_{\min}$ , and  $s_i(t) = 1$  then
7:       Add  $(i, j)$  to  $E_t^{rm}$  and assign  $w_{ij}^{rm}(t) = \exp[-d_{ij}(t)/\tau_{\text{edge}}]\kappa_{ij}(t)$ .
8:     end if
9:   end for
10: end for
```

At each excavation step a graph snapshot is defined as

$$G_t = (V_t^r \cup V_t^m, E_t^{rr} \cup E_t^{mm} \cup E_t^{rm}),$$

where V_t^r and V_t^m are the rock and TBM surface node sets defined in Section 3.1, E_t^{rr} denotes rock–rock voxel adjacency, E_t^{mm} denotes TBM surface adjacency, and E_t^{rm} denotes candidate rock–machine relations. Rock node features are

$$x_i^r(t) = [c_i, g_i, s_i(t)],$$

where $s_i(t)$ records whether voxel i remains active and eligible for interaction at step t . TBM node features are

$$x_j^m(t) = [p_j(t), n_j(t), \rho_j].$$

The intra-rock edge set E_t^{rr} is constructed from voxel neighbourhoods, and E_t^{mm} is constructed from local neighbourhoods on the parameterized TBM surface. The central graph-design decision is the construction of E_t^{rm} , the rock–machine candidate edges.

Edge-screening rules. A candidate edge (i, j) is included in E_t^{rm} only when four conditions are jointly satisfied:

$$\begin{cases} c_i \in \Omega_t, \\ d_{ij}(t) \leq \tau_{\text{edge}}, \\ \kappa_{ij}(t) \geq \eta_{\min}, \\ s_i(t) = 1, \end{cases}$$

namely active-zone membership, a distance threshold, surface-normal compatibility, and an excavation-state constraint. The first rule restricts candidate relations to the excavation-relevant rock volume Ω_t ; the second uses

$$d_{ij}(t) = \|c_i - p_j(t)\|$$

to keep only spatially close rock–machine pairs; the third uses the signed surface-normal compatibility

$$\kappa_{ij}(t) = \max\left(0, \frac{n_j(t)^\top (c_i - p_j(t))}{d_{ij}(t) + \epsilon}\right)$$

to keep only geometrically plausible facing relations; and the fourth prevents already excavated voxels from repeatedly contributing to later graph snapshots.

The active zone Ω_t separates face-contact plausibility from shield-side proximity within one active rock volume:

$$\Omega_t = \Omega_t^{\text{face}} \cup \Omega_t^{\text{shield}},$$

with, using cutterhead chainage $x_f(t)$, tail-shield chainage $x_{\text{tail}}(t)$, radial distance $r_i = \sqrt{y_i^2 + z_i^2}$, cutterhead radius R_c , shield radius R_s , face look-ahead length L_f , and radial active-zone buffer τ_{zone} ,

$$\begin{aligned}\Omega_t^{\text{face}} &= \{i \mid 0 \leq x_i - x_f(t) \leq L_f, r_i \leq R_c + \tau_{\text{zone}}\}, \\ \Omega_t^{\text{shield}} &= \{i \mid x_{\text{tail}}(t) \leq x_i \leq x_f(t), R_s \leq r_i \leq R_s + \tau_{\text{zone}}\}.\end{aligned}$$

Table 2 lists the parameter values used in all formal case runs. The radii and shield length come from the parameterized TBM geometry. The edge-screening parameters τ_{edge} and η_{min} are varied in Section 4.4. The active-zone parameters L_f and τ_{zone} define the main spatial support and are fixed in the reported formal runs.

Table 2: Active-zone and candidate-relation parameters used in all formal case runs.

Parameter	Value	Role
L_f	5.0 m	Face look-ahead length for Ω_t^{face}
τ_{zone}	5.0 m	Radial active-zone buffer around face and shield
R_c	4.0 m	Parameterized cutterhead radius
R_s	3.95 m	Parameterized shield radius
Shield length	10.0 m	Axial support for Ω_t^{shield}
τ_{edge}	2.0 m	Candidate-edge distance threshold in the main setting
η_{min}	0.3	Minimum surface-normal compatibility in the main setting
Surface sampling	0.5 m	TBM surface-node sampling resolution

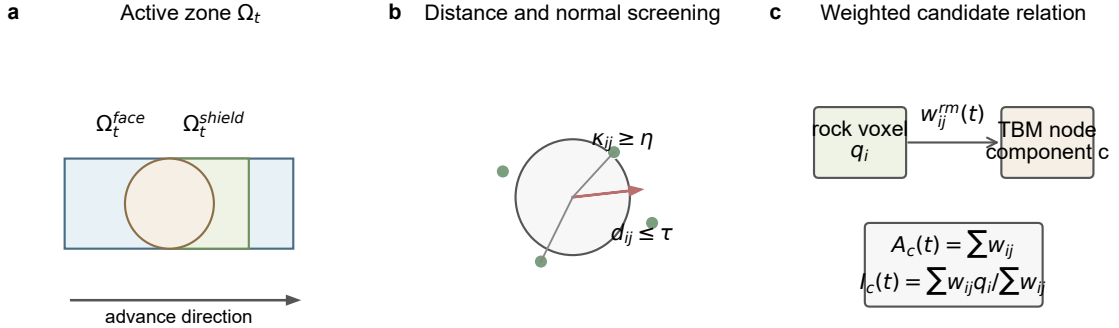


Figure 2: Geometry-constrained rock-machine candidate relation construction. Candidate relations are controlled by active-zone membership, distance decay, surface-normal compatibility, and excavation state. The construction defines spatially plausible relations; it does not recover measured contact.

Edge weight. For each candidate relation the geometry-compatibility weight is

$$w_{ij}^{rm}(t) = \exp\left(-\frac{d_{ij}(t)}{\tau_{\text{edge}}}\right) \kappa_{ij}(t).$$

The weight increases when a rock voxel is closer to the TBM surface and more compatible with the local surface normal. It is not a contact force or pressure. With V_t^r , V_t^m , E_t^{rr} , E_t^{mm} , E_t^{rm} , and $w_{ij}^{rm}(t)$ defined, the graph entities of Section 3.1 have become a computable rock-TBM interaction graph.

3.3 Component-indexed anomalous interaction field computation

Once the graph is constructed, the central graph computation is to aggregate screened rock-machine edges into a component-level anomalous interaction field. Response alignment and edge-level query are downstream readouts of this field rather than separate method branches.

Algorithm 2 Component-level anomalous interaction field computation

Require: Graph snapshots G_t , anomaly scores $q_i = \phi(g_i)$, component labels ρ_j

Ensure: Component $\times$ excavation-step field $\{I_c(t), \Delta I_c(t)\}$

- 1: **for** each excavation step t **do**
 - 2: **for** each TBM component c **do**
 - 3: Select rock-machine candidate edges $E_{c,t}^{rm} = \{(i, j) \in E_t^{rm} \mid \rho_j = c\}$.
 - 4: Compute geometric exposure $A_c(t) = \sum_{(i,j) \in E_{c,t}^{rm}} w_{ij}^{rm}(t)$.
 - 5: Compute anomaly intensity $I_c(t) = \sum_{(i,j) \in E_{c,t}^{rm}} w_{ij}^{rm}(t) q_i / (A_c(t) + \epsilon)$.
 - 6: Compute step change $\Delta I_c(t) = I_c(t) - I_c(t-1)$ when $t > 1$.
 - 7: **end for**
 - 8: **end for**
 - 9: Store edge-level contributions $s_{ij}(t) = w_{ij}^{rm}(t) q_i$ for subsequent query.
-

Component-level field. The geological contribution of each rock voxel is represented by a restrained TSP-derived anomaly score

$$q_i = \phi(g_i).$$

In the main formulation q_i is a bounded low-velocity anomaly score derived from the training active-zone distribution of P-wave velocity:

$$q_i = \text{clip} \left(\frac{Q_{95}^{\text{train}}(Vp) - Vp_i}{Q_{95}^{\text{train}}(Vp) - Q_5^{\text{train}}(Vp) + \epsilon}, 0, 1 \right).$$

This fixed training-reference scaling avoids per-chainage min-max normalisation and keeps field values comparable along chainage; lower V_p values correspond to higher geological anomaly scores under a common reference distribution.

For each TBM component c , let V_c^m denote the set of TBM surface nodes whose component label is c . Two component-level quantities are defined at each excavation step. The pure geometric exposure is

$$A_c(t) = \sum_{j \in V_c^m} \sum_{i: (i,j) \in E_t^{rm}} w_{ij}^{rm}(t),$$

which reports the geometric exposure of component c to the screened candidate rock-machine relation set. The geology-weighted spatial interaction intensity is

$$I_c(t) = \frac{\sum_{j \in V_c^m} \sum_{i: (i,j) \in E_t^{rm}} w_{ij}^{rm}(t) q_i}{\sum_{j \in V_c^m} \sum_{i: (i,j) \in E_t^{rm}} w_{ij}^{rm}(t) + \epsilon},$$

which reports the weighted geological anomaly intensity in the screened candidate relations incident to component c . $I_c(t)$ is defined for cutterhead, front shield, middle shield, and tail shield at each chainage step. Together with its step-level change $\Delta I_c(t)$, it forms the component $\times$ excavation-step anomalous interaction field. $A_c(t)$ is retained as an auxiliary exposure readout, not as part of the anomalous interaction field itself. Here, the field denotes the component $\times$ excavation-step matrix of $I_c(t)$ and $\Delta I_c(t)$ values, whereas a field readout denotes a downstream use of that matrix through response alignment or edge-contribution query.

Step-level tracking. Beyond the instantaneous intensity $I_c(t)$, the field also supports a step-level change view

$$\Delta I_c(t) = I_c(t) - I_c(t-1),$$

which reports how the anomalous interaction intensity around component c changes between consecutive excavation steps. Tracking $I_c(t)$ and $\Delta I_c(t)$ over the test chainage interval gives a component-indexed anomaly trajectory that indicates where and when the rock-machine interaction intensity rises or falls along the alignment.

The computed field is subsequently aligned with monitoring residuals for response readout and can be queried back to edge contributions. For response alignment, short-horizon TBM responses are evaluated through persistence residuals rather than raw responses:

$$e_{t+h}^{(k)} = r_{t+h}^{(k)} - r_t^{(k)},$$

where h is the diagnostic horizon and k indexes a response variable. The diagnostic question is whether component-indexed field readouts co-vary with the response change left after persistence has been removed:

$$I_c(t) \rightarrow e_{t+h}^{(k)}.$$

The reported association statistic is the Spearman correlation between $I_c(t)$ and $e_{t+h}^{(k)}$ over the evaluated test chainage interval. This readout links the graph-computed field to TBM response change; it is interpreted as field-residual co-variation within the evaluated chainage interval, not as causal attribution. The choice of diagnostic pairs, null variants, detrending, and robustness checks is part of the case evaluation design and is specified in Section 4.3. For edge-level traceability, candidate edges for any fixed diagnostic sample are ranked by their contribution

$$w_{ij}^{rm}(t) q_i,$$

and each contributing edge retains the rock voxel identity, the TBM surface node identity, the component label ρ_j , the distance $d_{ij}(t)$, the normal compatibility $\kappa_{ij}(t)$, the geometry weight $w_{ij}^{rm}(t)$, the anomaly score q_i , and the weighted contribution. The graph therefore not only produces a component-indexed field but also traces that field back to specific rock-machine spatial relations.

4 Case data and evaluation design

4.1 Tunnel cases and data alignment

The experiments use two real tunnel cases with complementary evidential roles. The BSLI (Baoshan-Shidian Longitudinal) tunnel case starts from DyK1017+205 and is evaluated under two diagnostic horizons: $h=1$ and $h=3$. The BSLI section is excavated in a mixed hard-rock environment dominated by moderately to strongly weathered granite and granitic gneiss, with local fault-fracture zones and intermittent low-velocity intervals that produce spatially variable V_p along the alignment. It provides a compact case for testing whether component-indexed field readouts are consistently constructed and related to response residuals over short target intervals. The SJLS (Shijiang-Luoshan) tunnel case starts from Dyk1252+411 and is evaluated under $h=3$. The SJLS section crosses a more clearly delineated low-velocity weak zone associated with fractured and water-bearing strata, producing a sustained external TSP velocity contrast over a longer test interval for field-residual association.

For each excavation step, the monitoring vector contains advance rate, cutterhead revolutions per minute (RPM), torque, thrust, penetration, and shield pressure. The diagnostic responses are advance rate, torque, thrust, penetration, and shield pressure. Raw monitoring records are collected at irregular per-ring intervals and are mileage-sorted and aggregated to

a uniform 1.0 m chainage grid by averaging records falling in the same rounded chainage bin. This aligns all response variables to the same chainage-indexed step t used by the graph sequence and avoids temporal misalignment between TBM monitoring and TSP chainage. No additional automated outlier deletion or winsorisation is applied in the reported diagnostic runs; the gridded monitoring channels are then standardized with means and standard deviations fitted on the training monitoring rows only. The diagnostic horizon h is applied in chainage space: $e_{t+h}^{(k)} = r_{t+h}^{(k)} - r_t^{(k)}$ uses the standardized gridded response at the target chainage, not the raw per-ring record.

The geological input is a TSP-derived voxel field. For BSLL, the project TSP voxel field is generated by the TSP203 system with a longitudinal extent of approximately 120 m around DyK1017+205, a cross-sectional extent of ± 20 m, and a nominal voxel resolution of $1.0 \text{ m} \times 1.0 \text{ m} \times 1.0 \text{ m}$. The voxel field stores P-wave velocity, S-wave velocity, density, dynamic Young’s modulus, velocity ratio, and Poisson’s ratio around the tunnel alignment. For SJLS, external P-wave and S-wave velocity profiles at Dyk1252+411 are first sampled at 1.0 m chainage intervals along the alignment and then converted into the same project TSP voxel schema by extruding each 1D velocity profile radially over the ± 20 m cross-sectional extent at $1.0 \text{ m} \times 1.0 \text{ m} \times 1.0 \text{ m}$ voxel resolution; the extruded voxels inherit the profile V_p and V_s values. Other schema fields such as velocity ratio, elastic moduli, Poisson’s ratio, or density are stored when available but are not used by the main graph-field computation. This profile-to-voxel conversion keeps the SJLS and BSLL inputs in a common voxel format while acknowledging that SJLS geological support is effectively 1D-extruded rather than fully 3D-inverted. The converted SJLS voxels are paired with excavation-step-aligned monitoring records.

4.2 Graph construction and field-readout settings

All cases use mileage-ordered partitioning. Component-level field readouts are computed from the last observed graph snapshot of each target sample, and the last observed response at step t is used to define the persistence residual for a target response at $t + h$. The formal case runs are summarised in Table 3.

Table 3: Formal case runs used in the field-readout experiments. Effective samples are chainage-indexed residual samples after applying the diagnostic horizon.

Case	Tunnel	Start chainage	Horizon	Train/Val./Test samples
BSLL h=1	BSLL	DyK1017+205	1	30 / 6 / 8
BSLL h=3	BSLL	DyK1017+205	3	29 / 6 / 7
SJLS h=3	SJLS	Dyk1252+411	3	76 / 16 / 17

The BSLL one-step run has target chainages 41–48 m in the test interval. The BSLL three-step run has target chainages 42–48 m. The SJLS run has target chainages 99–115 m. The graph construction uses $\tau_{\text{edge}} = 2.0 \text{ m}$, $\eta_{\text{min}} = 0.3$, and $\tau_{\text{zone}} = 5.0 \text{ m}$ in the main setting. The TBM surface is discretised into cutterhead, front-shield, middle-shield, and tail-shield surface nodes at a resolution of 0.5 m, with the per-case geometry summarised in Table 4. Rock nodes are the active, not-yet-excavated TSP voxels selected at each graph step by the active-zone formula in Section 3.2; under the common 1.0 m voxel grid, $\tau_{\text{zone}} = 5.0 \text{ m}$, and the fixed TBM template, this yields 3558 active rock nodes in the standardized support used for the reported graph statistics.

All formal runs use a 0.5 m surface sampling resolution. The stronger SJLS field-residual association is therefore not a by-product of coarser TBM surface discretisation.

TSP preprocessing follows a common voxel schema. Coordinates are converted to a local excavation coordinate by subtracting the minimum longitudinal coordinate, so the local X

Table 4: TBM geometry and surface sampling settings used in graph construction.

Case	Cutterhead radius (m)	Shield radius (m)	Front (m)	Middle (m)	Tail (m)	TBM nodes
BSLL h=1	4.0	3.95	3.0	3.5	3.5	1352
BSLL h=3	4.0	3.95	3.0	3.5	3.5	1352
SJLS h=3	4.0	3.95	3.0	3.5	3.5	1352

axis is aligned with advance direction. Each voxel can store $[V_p, V_s, E, V_p/V_s, \nu, \rho]$ when these attributes are available, and available attributes are standardized using statistics fitted on the training active-zone margin. This multi-attribute schema keeps the BSLL and SJLS inputs in a common voxel format. The main spatial interaction field computation uses only the cross-case consistent P-wave velocity attribute. The geological anomaly score is therefore a $[0, 1]$ low-velocity anomaly score derived from the training active-zone reference distribution of V_p ; derived or case-specific attributes such as V_s , modulus, velocity ratio, Poisson’s ratio, and density are not used in the main field computation.

4.3 Residual readouts and null variants

The diagnostic field–residual pairs are specified before the null-model and sensitivity checks: BSLL h=1 uses front-shield $I_c(t)$ against advance-rate residuals, BSLL h=3 uses front-shield $I_c(t)$ against shield-pressure residuals, and SJLS h=3 uses cutterhead $I_c(t)$ against shield-pressure residuals. These pairs serve as representative diagnostic slices covering a compact one-step BSLL interval, a multi-step BSLL shield-pressure interval, and the external SJLS tunnel case. They are not re-selected during null comparison or sensitivity analysis; the full exploratory component–response matrix is reported separately in Appendix A.

Four field-readout null variants test whether the proposed field readout can be reduced to simpler summaries. The chainage-only variant uses target chainage as the readout. The global-Vp variant uses the mean low-velocity anomaly score over active rock voxels, without component-specific geometry. The distance-only variant uses $A_c(t)$ and therefore contains geometry exposure but no geological anomaly weighting. The uniform-edge variant uses the unweighted mean of q_i over candidate rock–machine edges incident to the fixed component; it preserves the active-zone screening and component membership but removes distance-normal geometry weights.

A separate component-label permutation diagnostic is reported. At each chainage step, component labels are independently shuffled $N = 1000$ times, and the value assigned to the fixed diagnostic component is used as the permuted field-readout series. Both the observed and permuted field-readout series are linearly detrended against chainage before Spearman correlation with the detrended residual. The test assumes that component labels are exchangeable at each chainage step under the null; the resulting p -value is the proportion of permutations whose absolute correlation is at least as large as the observed. This randomisation diagnostic does not correct for the three cases tested and is not a block-wise permutation test.

Two interval-level robustness checks are applied to each fixed diagnostic pair. First, both field-readout and residual series are linearly detrended against chainage, and Spearman correlation is recomputed on the detrended series. Linear detrending is used as a minimal trend-removal device because the evaluated intervals are short; it is not intended to model nonlinear geological or operational trends. Second, a circular-shift check rolls the residual sequence by all non-zero shifts and reports the proportion of shifted correlations whose absolute value is at least as large as the observed value. These values are descriptive diagnostics rather than confirmatory probabilities, especially for the compact BSLL intervals. The full component–response matrix in the appendix is exploratory and unadjusted for multiple comparisons; it is used to show the field–response landscape, not to select case-general laws. For the fixed pairs, descriptive percentile

bootstrap intervals are reported for detrended Spearman correlations.

Because the proposed output is a component $\times$ excavation-step field rather than a single static summary, an additional step-change readout is reported. The associations of $I_c(t)$, $\Delta I_c(t)$, and $\Delta e_{t+h}^{(k)}$ are compared for the same fixed diagnostic pairs. This check evaluates whether the field contains excavation-step change information; it is not used to reselect diagnostic components or responses.

4.4 Sensitivity, ablation, and traceability settings

To account for TSP–TBM chainage alignment uncertainty, the fixed diagnostic field-readout series is also shifted by $\delta_x \in \{-1, 0, +1\}$ m using linear interpolation along target chainage, and raw and detrended Spearman correlations are recomputed. This test does not replace a full geophysical registration analysis, but it checks whether the reported diagnostic pattern depends on exact metre-level alignment.

A threshold sensitivity analysis varies τ_{edge} and η_{min} to check whether the field–residual pattern is tied to one specific graph-construction setting. Anomaly-definition sensitivity is evaluated by replacing the main Vp-based score with Vs-low, Vp/Vs-low, and mean(Vp, Vs) low-anomaly variants.

To separate relation-support effects from global geological trends, a relation-support ablation is also reported for the fixed pairs. It compares the proposed screened candidate-edge field with variants that remove the normal-compatibility threshold, use distance-only edge weights, or pool the entire TBM surface without component indexing. These ablations are interpreted as representation diagnostics, not as calibrated prediction benchmarks.

For traceability, candidate edges for a fixed diagnostic sample are ranked by their contribution $w_{ij}^{rm}(t)q_i$, retaining the rock voxel identity, the TBM surface node identity, the component label, the distance, the normal compatibility, the geometry weight, the anomaly score, and the weighted contribution. Edge-contribution concentration metrics are reported to indicate whether the query is dominated by a small number of candidate edges or distributed across the screened relation set. The results of all diagnostic readout variants, robustness checks, and sensitivity settings are reported in Section 5, and are interpreted as excavation-step-local field–response co-variation rather than causal evidence or calibrated operational risk.

5 Results

The results are organised around the three functions of the method rather than around prediction accuracy alone. Section 5.1 first shows how the construction produces an explicit and queryable graph support under controlled spatial parameters. Section 5.2 then examines whether edge aggregation produces component-indexed anomalous interaction fields. Sections 5.3–5.4 evaluate whether selected field readouts co-vary with monitoring residuals beyond simple trend or pooled-exposure alternatives. The final sensitivity and traceability results state the boundary of the evidence by testing parameter dependence, attribute definition, component indexing, and edge-contribution concentration.

5.1 Geometry-constrained interaction graph construction

The constructed spatial relation object is the first result. For each case, TSP-derived rock voxels and parameterized TBM surface nodes are mapped to the same chainage-referenced coordinate system, and the candidate relation set E_t^{rm} is rebuilt at each excavation step. Figure 2 illustrates the construction rule: candidate relations are screened by active-zone membership, distance threshold, surface-normal compatibility, and excavation state.

Table 5 reports the standardized graph support used in all three formal runs. The same TBM template, active-zone geometry, surface sampling, and edge thresholds are used throughout, so

the table documents a common graph support rather than case-specific graph-size variation. Case differences enter through geological anomaly scores, target excavation steps, and monitoring residuals. The screened graph is sparse in a literal bipartite-graph sense: the 14,620 candidate rock-machine edges represent about 0.30% of the 3558×1352 possible rock-surface node pairs before geometric screening.

Table 5: Standardized graph support template applied in all formal runs. Edge shares are computed from candidate rock-machine relations under the common main graph setting. The identical counts document that case contrasts originate from geological attributes and residuals, not from graph-size variation.

Case	Rock nodes	TBM nodes	Candidate edges	Cutterhead share	Shield share
BSLL h=1	3558	1352	14620	0.145	0.855
BSLL h=3	3558	1352	14620	0.145	0.855
SJLS h=3	3558	1352	14620	0.145	0.855

The field statistics in Table 6 explain why SJLS provides clearer field-residual evidence under the same graph support template. The distinction does not come from graph size. It comes from the more uniformly high low-velocity anomaly field assigned to the standardized support. This interpretation is conditional on the SJLS geological support being a 1D-extruded TSP velocity profile rather than a fully 3D-inverted voxel field; the SJLS result is therefore evidence of longitudinal field-residual consistency under a common support, not proof of fully 3D component differentiation. Within that boundary, the SJLS interval provides a clearer field-residual pattern than the compact BSLL intervals.

A notable feature of Table 6 is that the candidate-edge mean anomaly and $I_c(t)$ are numerically very close in all three cases (e.g. BSLL h=1: 0.552 ± 0.157 vs. 0.552 ± 0.158). This near-equality is expected rather than degenerate: $I_c(t)$ aggregates q_i over screened candidate edges. Within the present field-readout pairs, the empirical distinction is mainly between component-indexed candidate-edge aggregation and pooled or global summaries; the additional numerical effect of distance-normal weighting is secondary. The value of the proposed field readout lies in retaining the component-specific, excavation-step-specific candidate-edge aggregation that the pooled active-zone mean conceals. The null-model comparison in Section 5.4 tests this directly by comparing the proposed field readout against the candidate-edge mean (uniform-edge variant) and the global active-zone mean (global-Vp variant).

Table 6: Attribute statistics of the anomalous interaction field. Mean anomaly is the candidate-edge mean of the bounded low-velocity score q_i ; $I_c(t)$ is summarised over component-level field values and test chainages.

Case	Mean anomaly score	$I_c(t)$	$A_c(t)$
BSLL h=1	0.552 ± 0.157	0.552 ± 0.158	1170.8 ± 296.9
BSLL h=3	0.537 ± 0.129	0.537 ± 0.130	1170.9 ± 297.4
SJLS h=3	0.913 ± 0.045	0.913 ± 0.045	1170.9 ± 294.2

Table 7 summarises the retained component-indexed field structure directly. BSLL shows stronger cross-component contrast in $I_c(t)$, while SJLS shows weaker cross-component contrast, as expected for a 1D-extruded velocity profile with common cross-sectional support. The two cases therefore emphasise different aspects of the field: BSLL illustrates component differentiation under 3D voxel support, whereas SJLS illustrates longitudinal field-residual co-variation.

5.2 Anomalous interaction field patterns

The main graph-derived output is the component $\times$ excavation-step field built from $I_c(t)$. In each case, cutterhead, front-shield, middle-shield, and tail-shield values are computed across

Table 7: Component contrast in the $I_c(t)$ field. The statistics summarize the per-component, per-chainage field matrix and are not response-model scores.

Case	Component-mean range	Mean within-component SD	Full $I_c(t)$ range
BSLL h=1	0.220	0.132	0.570
BSLL h=3	0.211	0.102	0.426
SJLS h=3	0.016	0.044	0.150

the test chainage interval as TSP anomaly summaries over screened candidate rock-machine relations.

Figure 3 previews the per-component $I_c(t)$ series across the three case runs on the common graph support. The BSLL cases operate over compact chainage intervals (about 6–7 m), while SJLS spans a longer test interval (about 16 m) with a markedly higher anomaly level. This figure motivates the case-level diagnostic contrast examined in the remainder of this section.

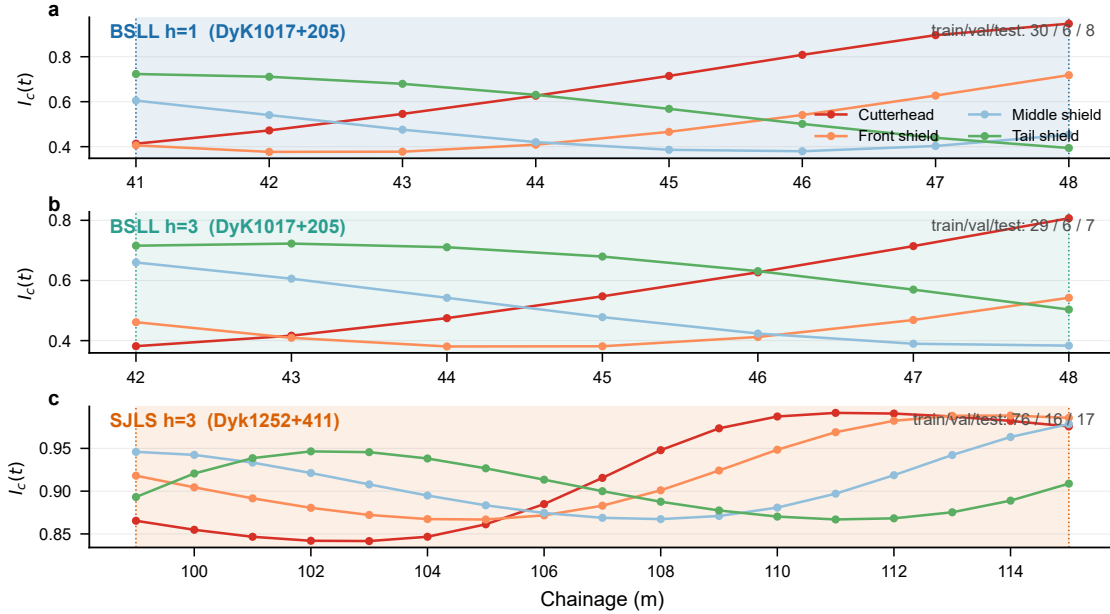


Figure 3: Per-component $I_c(t)$ field series for the three formal case runs on the common standardized graph support. Shaded bands mark the test chainage interval; BSLL cases are compact, while SJLS provides a longer test interval with a uniformly high low-velocity anomaly level.

Figure 4 links these field readouts to the fixed diagnostic residual traces. Each case panel holds the diagnostic component and response fixed, showing the selected $I_c(t)$ series above the corresponding persistence residual. This layout makes the evidence chain explicit: the graph first produces a component-indexed field readout, and the residual trace then tests whether that readout co-varies with response changes left after persistence.

5.3 Field-response residual associations

The reported association is the Spearman correlation between $I_c(t)$ and the persistence residual $e_{t+h}^{(k)}$. Table 8 reports the representative diagnostic pairs together with raw Spearman correlation, linearly detrended Spearman correlation, and the circular-shift diagnostic. The pairs are representative diagnostic slices selected from the main field-response matrix before the secondary null, alignment, and sensitivity checks. They cover the compact BSLL h=1 interval, the BSLL h=3 shield-pressure interval, and the external SJLS shield-pressure case, while retaining

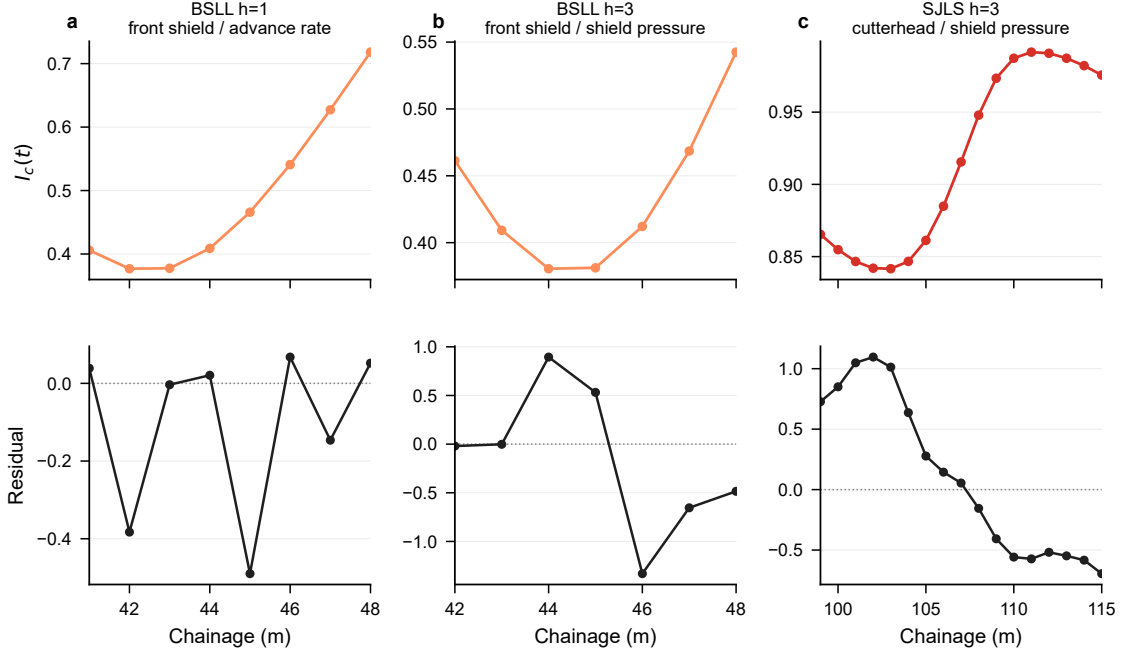


Figure 4: Fixed diagnostic field-residual traces. Upper panels show the selected component $I_c(t)$ series for each formal case; lower panels show the matching persistence residual series in the standardized monitoring scale used for residual computation.

component-response combinations with direct engineering plausibility. They are not claimed to be unique physically responsible components or globally strongest pairs; Appendix A reports the full exploratory matrix without multiplicity adjustment. The values are descriptive and are not treated as confirmatory hypothesis tests.

Table 8: Representative diagnostic field-residual diagnostics for fixed component-response pairs. Bootstrap intervals are descriptive percentile intervals for detrended Spearman correlations (10,000 resamples).

Case	Diagnostic pair	n	Raw ρ	Detrended ρ	95% CI	Shift check
BSLL h=1	Front shield-advance rate	8	0.381	0.214	$[-0.671, 1.000]$	0.375
BSLL h=3	Front shield-shield pressure	7	-0.750	-0.464	$[-0.957, 0.769]$	0.429
SJLS h=3	Cutterhead-shield pressure	17	-0.912	-0.772	$[-0.968, -0.375]$	0.059

The bootstrap intervals are wide for the two BSLL cases, reflecting the compact test intervals. SJLS shows the clearest field-residual co-variation: its cutterhead $I_c(t)$ -shield-pressure residual association remains strong after linear detrending ($\rho = -0.772$), whereas the BSLL evidence weakens under detrending. This difference is consistent with the longer SJLS test interval and clearer external TSP velocity contrast. The BSLL results are retained as compact diagnostic checks within their evaluated intervals.

Table 9 adds the step-change readout of the component $\times$ excavation-step field. The clearest additional pattern is observed in SJLS: although $\Delta I_c(t)$ is weakly associated with the residual level, it has a large negative association with the step-level residual change ($\rho = -0.724$ raw and -0.732 after detrending). This supports reading the field as an excavation-step anomaly trajectory, not only as a static low-velocity summary. The BSLL step-change values are reported as descriptive checks over six or seven differenced samples.

Appendix A reports the full component-response Spearman matrix for the field readout $I_c(t)$. The matrix is exploratory and unadjusted for multiple testing. The fixed pairs in Table 8

Table 9: Step-change association for the component-level anomalous interaction field. Values are Spearman correlations for the same fixed diagnostic pairs as Table 8.

Case	Comparison	n	Raw ρ	Detrended ρ
BSLL h=1	$I_c(t)$ vs. residual	8	0.381	0.214
BSLL h=1	$\Delta I_c(t)$ vs. residual	7	0.429	-0.143
BSLL h=1	$\Delta I_c(t)$ vs. Δ residual	7	0.214	-0.214
BSLL h=3	$I_c(t)$ vs. residual	7	-0.750	-0.464
BSLL h=3	$\Delta I_c(t)$ vs. residual	6	-0.600	-0.600
BSLL h=3	$\Delta I_c(t)$ vs. Δ residual	6	0.029	-0.543
SJLS h=3	$I_c(t)$ vs. residual	17	-0.912	-0.772
SJLS h=3	$\Delta I_c(t)$ vs. residual	16	-0.071	-0.474
SJLS h=3	$\Delta I_c(t)$ vs. Δ residual	16	-0.724	-0.732

are diagnostic slices interpreted together with the null, detrending, sensitivity, and traceability checks below.

5.4 Null and relation-support comparisons

The null-model comparison asks whether simpler summaries account for the fixed diagnostic associations. Figure 5 shows raw and detrended correlations for the proposed field readout and four field-readout null variants. The detrended comparison is the main diagnostic because chainage and global anomaly can both carry strong monotonic trends in short test intervals. Table 10 lists the detrended correlations for the proposed field readout and four field-readout null variants.

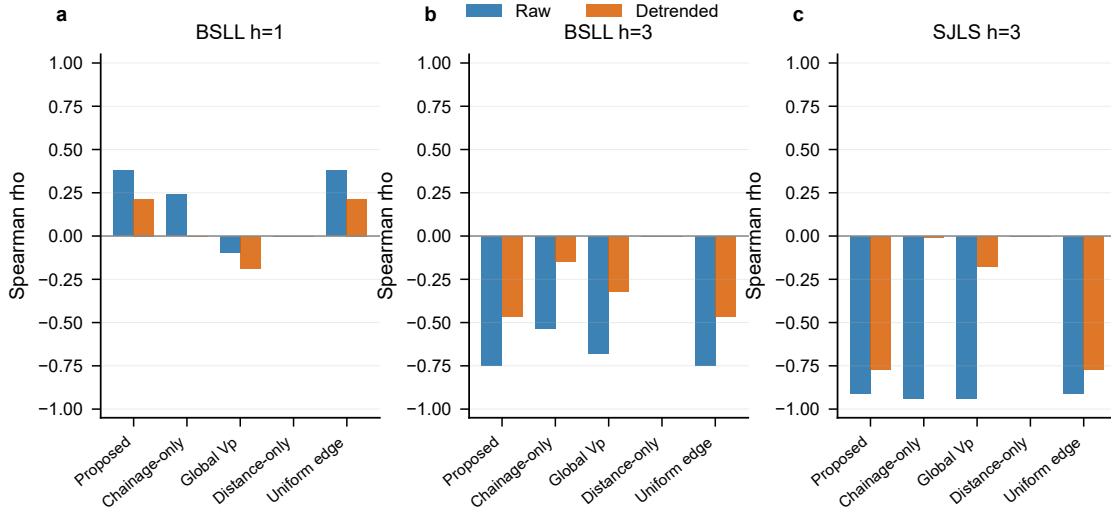


Figure 5: Field-readout null-model comparison for the representative diagnostic pairs. Bars report raw and linearly detrended Spearman correlations for the proposed field readout and four readout variants. For the fixed pairs, the uniform-edge readout is essentially the same as the proposed readout; the figure therefore separates the component-indexed anomaly aggregation from any additional performance effect of distance-normal geometry weighting. The comparison evaluates field-readout diagnostic structure, not prediction accuracy.

For SJLS, the raw chainage-only and global-Vp associations are high, but both drop markedly after detrending. The proposed field readout retains a strong detrended association, indicating that the component-specific candidate-edge aggregation retains residual-consistent structure beyond a simple chainage trend. Because the uniform-edge and geometry-weighted readouts are nearly identical in these fixed pairs, the comparison supports component-indexed candidate-edge

Table 10: Detrended Spearman correlations for the proposed field readout and four field-readout null variants. The component-label disruption diagnostic is reported separately in Table 12.

Case	Proposed	Chainage	Global Vp	Distance only	Uniform edge
BSLL h=1	0.214	0.000	-0.190	0.000	0.214
BSLL h=3	-0.464	-0.145	-0.321	0.000	-0.464
SJLS h=3	-0.772	-0.009	-0.179	0.000	-0.772

aggregation more directly than a separate gain from distance-normal geometry weighting.

Table 11 further separates relation-support effects. Removing the normal-compatibility threshold has little effect on the fixed-pair correlations in these samples, and the distance-weighted anomaly variant weakens but retains the SJLS association. The most important contrast is the pooled TBM surface variant: after removing component indexing, the detrended SJLS association changes from -0.772 to 0.265 , and the BSLL h=3 association is also weakened. These comparisons support component-indexed candidate-edge aggregation more strongly than an independent empirical gain from the distance-normal weight itself. Here, “distance-weighted anomaly” denotes a geology-weighted field built with distance weights but without normal compatibility; it differs from the distance-only exposure null in Table 10, which contains geometry exposure but no anomaly score.

Table 11: Relation-support ablation for fixed diagnostic pairs. Values are Spearman correlations between each ablated field readout and the fixed monitoring residual.

Case	Variant	Raw ρ	Detrended ρ
BSLL h=1	Proposed	0.381	0.214
BSLL h=1	No normal screen	0.381	0.214
BSLL h=1	Distance-weighted anomaly	0.381	0.214
BSLL h=1	Pooled TBM surface	0.452	0.167
BSLL h=3	Proposed	-0.750	-0.464
BSLL h=3	No normal screen	-0.750	-0.464
BSLL h=3	Distance-weighted anomaly	-0.750	-0.464
BSLL h=3	Pooled TBM surface	0.536	-0.179
SJLS h=3	Proposed	-0.912	-0.772
SJLS h=3	No normal screen	-0.912	-0.772
SJLS h=3	Distance-weighted anomaly	-0.860	-0.586
SJLS h=3	Pooled TBM surface	-0.488	0.265

A step-wise component-label disruption diagnostic ($N = 1000$ permutations, with component labels independently shuffled at each chainage step and both the observed and disrupted series linearly detrended against chainage before correlation) evaluates whether the fixed component label matters after the common graph support and chainage trend are removed. As reported in Table 12, the descriptive permutation proportions are 0.588 (BSLL h=1), 0.325 (BSLL h=3), and < 0.001 (SJLS h=3). This diagnostic is a label-disruption check rather than a block-wise or case-general permutation test. Because relabelling is performed independently at each chainage step, the test can overstate component specificity and does not preserve complete component series. The values are read only as descriptive disruption diagnostics.

Table 13 reports a stricter whole-series component reassignment diagnostic. Instead of shuffling labels at each chainage, it compares the target component’s complete detrended $I_c(t)$ series with the complete series from the other components for the same residual. This check motivates a cautious component-specific interpretation: the fixed target is not always the strongest component series, especially for SJLS where tail-shield has a larger absolute detrended association. The result supports field-residual consistency more strongly than unique component-label specificity. SJLS is therefore treated as the strongest longitudinal residual-consistency case, whereas BSLL remains the only case with fully 3D TSP voxel support for component differentiation, although its short test interval limits residual evidence.

Table 12: Step-wise component-label disruption diagnostic on detrended series. At each chainage step, component labels are independently shuffled ($N = 1000$ permutations); both observed and permuted field-readout series are linearly detrended against chainage before computing Spearman correlation with the detrended residual. The reported proportion is descriptive and unadjusted.

Case	Component	n	Observed ρ	Permutation p
BSLL h=1	front shield	8	0.214	0.588
BSLL h=3	front shield	7	-0.464	0.325
SJLS h=3	cutterhead	17	-0.772	< 0.001

Table 13: Whole-series component reassignment diagnostic. Values are detrended Spearman correlations between each complete component $I_c(t)$ series and the fixed residual; no per-step label shuffling is used.

Case	Target component	Target ρ	Other ρ range	Strongest other ρ	Interpretation
BSLL h=1	front shield	0.214	-0.214 to 0.167	-0.214	tied absolute strength
BSLL h=3	front shield	-0.464	-0.500 to 0.464	-0.500	target not strongest
SJLS h=3	cutterhead	-0.772	-0.135 to 0.877	0.877	target not strongest

The null variants separate three questions. The global-Vp variant asks whether a case-level active-zone mean is sufficient. The distance-only variant asks whether geometric exposure alone accounts for the residual association. The uniform-edge variant keeps the fixed component and the screened candidate-edge set, but removes distance-normal weights by using an unweighted mean of q_i . In Table 10, the uniform-edge values are numerically the same as the proposed field readout for the fixed pairs. The comparisons therefore support component-specific candidate-edge aggregation over global active-zone averaging. Within these samples, the incremental empirical benefit of geometry weighting remains secondary. The geometry weights remain part of the representation because they define a transparent, queryable edge contribution $w_{ij}^{rm}(t)q_i$, but their independent empirical gain requires larger or more spatially heterogeneous cases.

5.5 Sensitivity to graph and anomaly settings

Figure 6 tests whether the fixed $I_c(t)$ -residual pairs are preserved when the edge-distance threshold τ_{edge} and normal-compatibility threshold η_{min} are varied. The same field-response pair is held fixed in each case, so the figure evaluates threshold stability rather than searching for a new strongest pair.

Table 14 reports the anomaly-definition sensitivity for the same fixed diagnostic pairs. SJLS remains strong under Vp, Vs, and mean(Vp, Vs) low-anomaly definitions after detrending. BSLL is more sensitive to the anomaly definition, which is consistent with its compact test intervals.

Table 14: Detrended Spearman correlations under alternative anomaly-score definitions for the fixed diagnostic pairs.

Case	Vp main	Vs low	Vp/Vs low	Mean(Vp, Vs)
BSLL h=1	0.214	-0.214	0.214	-0.024
BSLL h=3	-0.464	0.179	-0.179	-0.643
SJLS h=3	-0.772	-0.873	0.338	-0.863

Table 15 reports the TSP-TBM chainage alignment uncertainty check. The SJLS association remains negative under ± 1 m offsets and remains strong after detrending at the 0 and +1 m settings. The BSLL checks are more variable, again reflecting their short test intervals.

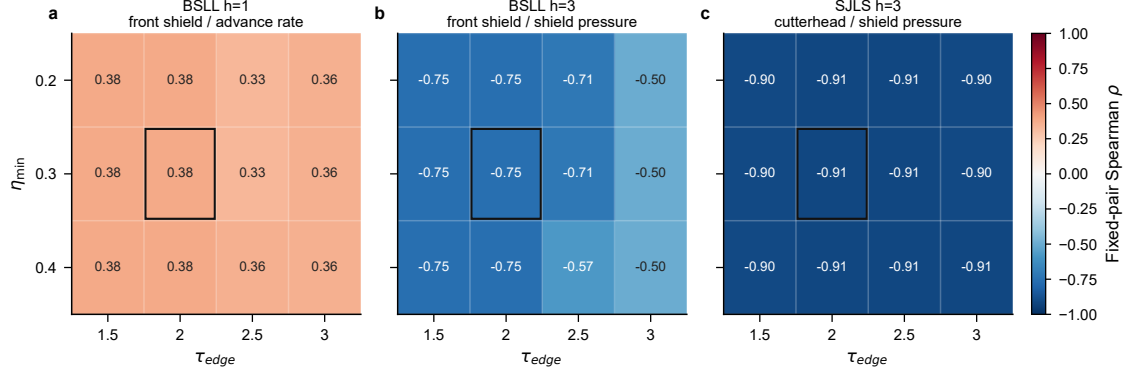


Figure 6: Sensitivity of fixed $I_c(t)$ -residual pairs to graph-construction thresholds. Each panel reports raw Spearman ρ while varying τ_{edge} and η_{min} for one formal case; the field-response pair is held fixed throughout.

Table 15: Sensitivity of fixed diagnostic pairs to ± 1 m longitudinal field-readout alignment offsets. Values report raw/detrended Spearman correlations.

Case	-1 m	0 m	+1 m
BSLL h=1	0.250 / -0.107	0.381 / 0.214	-0.071 / 0.250
BSLL h=3	-0.143 / 0.257	-0.750 / -0.464	-0.657 / -0.714
SJLS h=3	-0.850 / -0.397	-0.912 / -0.772	-0.938 / -0.874

5.6 Edge-contribution traceability

The graph representation also supports edge-level inspection. For the SJLS diagnostic pair, Figure 7 shows the top contributing candidate rock-machine relations at the selected chainage. Each contribution is traceable to a rock voxel, a TBM surface node, a distance, a normal compatibility value, a geometry weight, and a low-velocity anomaly score; the figure displays the leading spatial edges and their ranked contributions, while the full edge-attribute rows remain available in the diagnostic outputs.

Table 16 summarises the concentration of weighted edge contributions across the fixed diagnostic component in each case. The top five edges account for only about 0.5% of the total contribution and the normalised entropy is close to one, indicating that the field readout is distributed across a broad screened relation set rather than dominated by a few extreme edges. Edge-level query should therefore be read as a way to inspect contributing relations and their ranks, not as evidence that one or two edges physically caused the monitoring response.

Table 16: Mean edge-contribution concentration across the fixed diagnostic component in the test interval. Effective edge number is computed from the normalised edge-contribution weights.

Case	Top-5 share	Top-10 share	Effective edges	Entropy
BSLL h=1	0.005	0.010	3033.2	0.985
BSLL h=3	0.005	0.010	3024.7	0.985
SJLS h=3	0.005	0.009	1890.9	0.992

This traceability is the main practical difference between the proposed graph field readout and a monitoring-only residual curve. The field readout can be traced back to the spatial relation set that produced it, whereas the residual curve alone cannot identify which geological voxels and machine components were involved.

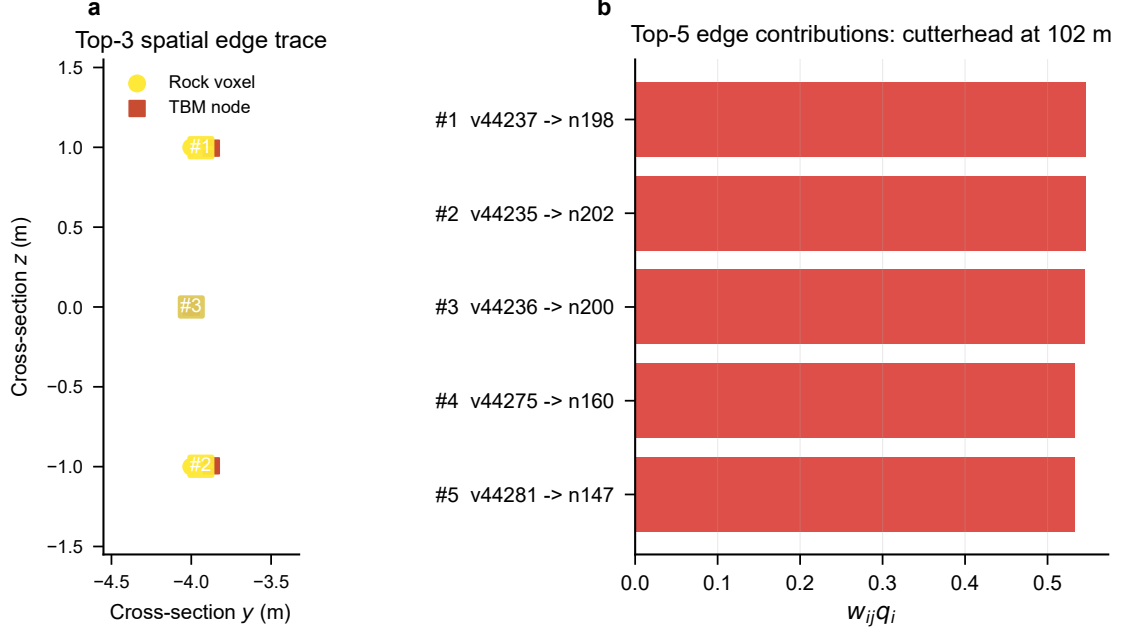


Figure 7: Traceability example for a fixed diagnostic pair. Candidate rock-machine edges are ranked by $w_{ij}^{rm}(t)q_i$; the left panel maps the top three spatial edges in cross-section, and the right panel reports the top five weighted contributions.

6 Discussion

6.1 Spatial representation contribution

The main contribution is an excavation-step-indexed spatial relation model, not a higher-accuracy response predictor. The model constructs a component $\times$ excavation-step anomalous interaction field from a TSP-derived rock voxel field and an advancing parameterized TBM surface, then aligns this field with monitoring responses. Monitoring sequences describe response inertia, and TSP voxels describe geological heterogeneity, but neither representation alone states which geological voxels are candidate interaction partners for which TBM components at a given chainage. The proposed graph closes this representational gap by linking rock voxels, TBM surface nodes, and geometry-compatible candidate edges in one excavation coordinate system, and by aligning the resulting field with monitoring residuals to support excavation anomaly discovery and jamming-risk-oriented interpretation.

For spatial modelling, the graph is an explicit data representation rather than a predictive architecture. It has defined entities, relation types, screening rules, edge weights, component aggregation, and traceability back to individual candidate relations. The field results show that this representation organises rock-machine relations into component-indexed field readouts.

From a digital-earth perspective, the contribution is an underground spatial data modelling layer for an advancing construction process. It organises static geological support and advancing TBM surface geometry into excavation-step-indexed spatial relations that can be inspected and queried, while monitoring records provide aligned response readouts. The layer is compatible with digital twin workflows, but it addresses a narrower representational problem: how to make the rock-machine relation support explicit before any prediction, control, or risk-calibration module is built.

6.2 Diagnostic readout beyond monitoring curves

Comparing $I_c(t)$ with persistence residuals reframes the evaluation question from “does the graph beat persistence?” to “does the graph encode spatial diagnostic information consistent with the response changes left after persistence?” This framing matches the present data scale. The SJLS results illustrate the value of this framing: raw chainage and global Vp summaries are substantially associated with the selected residual, but their associations largely disappear after linear detrending, whereas the proposed candidate-edge field readout remains strong. The step-change readout further shows a large negative association between $\Delta I_c(t)$ and the step-level shield-pressure residual change in SJLS, supporting the interpretation of the output as an excavation-step anomaly field rather than only a static geological summary.

The BSLL results show that the graph and field readouts are constructed consistently in compact intervals, while residual-consistency evidence is more sensitive to detrending and anomaly definition. This contrast defines the claim boundary: the evidence supports a spatial field-representation claim, not a universal component–response law.

6.3 Evidence scope across tunnel cases

The contrast between SJLS and BSLL is not treated as a success–failure dichotomy. It indicates that field-readout usefulness depends on geological contrast, monitoring quality, operational stability, and the length of the evaluated chainage interval. SJLS contains a longer test interval and a clearer external TSP velocity contrast, which makes field–residual co-variation more visible. The BSLL intervals are compact and may be more strongly affected by operational control and local construction variability.

The circular-shift checks are also read conservatively. They are useful as interval-level diagnostics for monotonic sequence structure, while formal multiplicity-adjusted testing would require broader evidence. The most coherent support comes from the combined pattern: fixed diagnostic pairs, detrended comparison, field-readout null models, threshold sensitivity, anomaly-definition sensitivity, and edge-level traceability. The relation-support ablation adds a more specific boundary: component pooling weakens the fixed diagnostic associations, whereas removing the normal-compatibility threshold or distance–normal weighting has little effect in the current samples. The supported claim is therefore centred on component-indexed candidate-edge aggregation rather than a separate performance gain from geometry weighting alone.

The step-wise component-label disruption diagnostic (Table 12) is retained only as a label-disruption check. Because it shuffles labels independently at each chainage step, it does not preserve complete component trajectories and should not be read as proof that physically different TBM components are exchangeable. The whole-series component reassignment diagnostic (Table 13) is more conservative and shows that the fixed target component is not always the strongest complete component series. The evidence therefore supports candidate-edge field consistency more strongly than unique component-label specificity.

6.4 Interpretation boundaries

The proposed field readouts are summaries of screened TSP-derived anomalies around TBM components, with geometry weights retained for edge-level traceability. They are not measured contact force, contact pressure, jamming probability, causal attribution, or calibrated operational risk. This boundary follows from the graph definition: it encodes candidate spatial relations, not observed physical contact. Independent engineering labels would be required before the field readouts could be interpreted as risk probabilities or contact-force estimates. The edge-contribution concentration results also show that the field readout is distributed across many screened candidate relations. Edge-level query should therefore be interpreted as ranked spatial traceability, not as identification of a small set of physically causal contact edges.

A second boundary concerns the standardized graph support. All three formal runs use the same parameterized TBM surface, active-zone geometry, surface sampling resolution, and edge thresholds, so the per-step node and edge counts are the same across cases by construction. This standardization keeps the field readout comparable across cases and prevents graph-size variation from being confounded with geological contrast, but it also means that the current framework cannot test whether a different TBM geometry, a coarser surface sampling, or a different active-zone extent would change the diagnostic pattern. The standardized support is therefore both a comparability device and a scope limitation: the reported evidence is conditional on one graph template, and sensitivity to machine geometry and active-zone dimensions remains for future work.

6.5 Limitations and future work

Several limitations remain. First, the evidence is interval-specific: it comes from two tunnel cases and their evaluated intervals. The reported associations are therefore descriptive diagnostics rather than case-general component–response laws. Second, TSP inversion uncertainty is not propagated through the voxel graph; future work should attach uncertainty to voxel attributes and evaluate how inversion and spatial resolution affect candidate relation sets. For the SJLS case in particular, the geological support is a 1D-extruded velocity profile rather than a fully 3D-inverted voxel field, so cross-sectional geological variation is not captured by the current field readout.

Third, the step-wise component-label disruption diagnostic only disrupts labels within the standardized graph support. Larger datasets would allow stronger null designs and inference that accounts for multiple comparisons. Fourth, the study lacks independent engineering labels such as jamming records, shield pressure anomaly logs, construction intervention logs, or geological exposure records. Such evidence would be required before the field readouts could be evaluated as operational risk indicators. Finally, the main field readout uses a transparent Vp-based low-velocity anomaly score. Multi-attribute scores using Vs, modulus, Vp/Vs, Poisson’s ratio, or density require explicit weighting rules or independent validation before they can replace the main field readout.

7 Conclusion

This study builds a geometry-constrained rock–TBM spatial interaction graph from three inputs: a TSP-derived rock voxel field, an advancing parameterized TBM surface, and excavation-step-aligned monitoring records. The core method constructs this graph at each excavation step and aggregates weighted candidate edges by TBM component into a component $\times$ excavation-step anomalous interaction field. The resulting field contains $I_c(t)$ and $\Delta I_c(t)$, can be aligned with monitoring responses, and retains traceable edge contributions for excavation anomaly indication, component-indexed readout, and jamming-risk-oriented interpretation.

The two case studies indicate that the anomalous interaction field organises rock–machine relations into interpretable component-indexed excavation-step patterns and, most clearly in the SJLS case, retains residual-consistent co-variation after simple trend-based null explanations are considered. The component-level field also provides a step-change readout. In the SJLS case, $\Delta I_c(t)$ has a large negative association with the step-level shield-pressure residual change. Relation-support ablations indicate that, within the present field-readout pairs, the empirical distinction is mainly between component-indexed candidate-edge aggregation and pooled or global summaries; the additional numerical effect of distance-normal weighting is secondary. The evidence supports a spatial field-representation claim rather than contact-force recovery, calibrated jamming risk, unique component attribution, or broad predictive dominance.

The proposed field provides a spatial representation basis for future rock–machine interaction

analysis and digital-twin-oriented TBM excavation analysis. In particular, the component $\times$ excavation-step anomalous interaction field can be queried alongside monitoring responses to identify which TBM component is exposed to anomalous rock conditions at which chainage. It therefore offers a structured input for excavation anomaly indication and jamming-risk-oriented interpretation in future excavation-analysis workflows.

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Disclosure statement

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Data availability statement

The raw TBM monitoring and TSP data are project data and are not publicly available because they contain engineering and site information. The materials that can be shared are the experiment configurations, analysis scripts, figure scripts, and derived result summaries used to reproduce the reported tables and figures. These materials are available from the corresponding author upon reasonable request, subject to project data-use permissions; the raw site data are not redistributed. To support reproducibility of the spatial graph construction without disclosing site data, the authors can also provide a synthetic demonstration package containing a small voxel field, TBM surface nodes, candidate edge lists, field-calculation scripts, and residual diagnostic scripts.

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A Exploratory field–response matrix

Table 17 reports the full unadjusted component–response Spearman matrix for all cases, included here for completeness. These values are exploratory field-readout diagnostics and are not adjusted for multiple testing.

Table 17: Full unadjusted component–response Spearman matrix for $I_c(t)$ –persistence-residual association. Correlations are exploratory field-readout diagnostics and are not adjusted for multiple testing.

Case	Component	Adv. rate	Torque	Thrust	Penetr.	Shield press.
BSLL h=1	Cutterhead	0.238	-0.286	0.048	-0.095	0.095
BSLL h=1	Front shield	0.381	-0.429	0.190	0.095	-0.095
BSLL h=1	Middle shield	-0.048	0.262	0.452	-0.167	0.143
BSLL h=1	Tail shield	-0.238	0.286	-0.048	0.095	-0.095
BSLL h=3	Cutterhead	-0.214	-0.714	-0.036	-0.214	-0.536
BSLL h=3	Front shield	0.286	-0.179	0.143	0.250	-0.750
BSLL h=3	Middle shield	0.214	0.714	0.036	0.214	0.536
BSLL h=3	Tail shield	0.179	0.750	0.107	0.143	0.571
SJLS h=3	Cutterhead	0.860	0.446	-0.669	0.669	-0.912
SJLS h=3	Front shield	0.640	0.152	-0.350	0.475	-0.745
SJLS h=3	Middle shield	-0.225	-0.449	0.478	-0.282	-0.042
SJLS h=3	Tail shield	-0.846	-0.324	0.733	-0.811	0.784